Worksheet

Power Series, Taylor Series

For each of the following power series

a. determine the radius of convergence
b. determine interval of absolute convergence
c. if the radius of convergence is finite, determine whether the power series converges at the end points of the interval of absolute convergence

1. $\displaystyle\sum_{k=0}^{\infty} \frac{k+1}{k+2} x^k$

2. $\displaystyle\sum_{k=0}^{\infty} \frac{x^k}{k4^k}$

3. $\displaystyle\sum_{k=0}^{\infty} \frac{x^{2k}}{k4^k}$

4. $\displaystyle\sum_{k=0}^{\infty} \frac{(-2)^k x^k}{k!}$

5. $\displaystyle\sum_{k=0}^{\infty} \frac{(x-1)^k}{k^2 3^k}$

6. $\displaystyle\sum_{k=0}^{\infty} \frac{(2x+3)^k}{k+1}$

For each of the following functions find the Taylor's series expansion for the function (centered at *c*)

1. $f(x) = xe^{-x}$
$c = 0$

2. $f(x) = x^2 \ln(1+x)$
$c = 0$

3. $f(x) = \dfrac{1}{1+4x}$
$c = 0$

4. $f(x) = \dfrac{x - \sin(x)}{x^3}$
$c = 0$

5. $f(x) = x \ln x$
$c = 1$

For each of the following functions find the first four non-zero terms in the Taylor's series expansion for the functioned (centered at c)

1. $f(x) = \frac{1}{x}$
   $c = 2$

2. $f(x) = \cos x$
   $c = \pi$

3. $f(x) = \frac{x}{1-x}$
   $c = 0$

For each of the following functions find an approximation for the value of the function at the given point x_0 for which the approximation error is less that 10^{-6}

1. $f(x) = \cos x$
   $x_0 = 0.2$

2. $f(x) = \arctan x^2$
   $x_0 = 0.2$

3. $f(x) = \mathrm{e}^x$
   $x_0 = -0.2$

4. $f(x) = e^x$
   $x_0 = 0.2$